

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mock Final  
Duration: 1 hour 40 minutes

Semester: Summer26  
Full Marks: 40

**CSE 340: Computer Architecture**

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. a. **Convert**  $-111011.11011 \times 2^{12}$  in 26-bit IEEE-754 format where, 3
- i. Size of the fraction field is 13 bits.
- ii. Size of the fraction field is 17 bits.
- Show** the equivalent Hex representation of your conversions.
- iii. Suppose you plan to use the converted number in a subsequent calculation where precision is crucial. Given a choice between the two IEEE formats (with 13-bit and 17-bit fraction fields), which format would provide a more accurate conversion for your calculation? Justify
- CO1
- b. **Subtract** -6.55 from 15.21 using IEEE-754 single-precision floating-point representation. 2
- Consider** 5 decimal digits when you are converting from decimal to binary.
- c. For a 20-bit register with 1 bit for signs, 6 bits for exponents, and the rest for fractions, what will be the actual exponent range of the numbers that can be represented using IEEE 754 floating point representations? 2
2. a. Consider the below code sequence. Assume  $X_1$  has the value 0x00001232 and  $X_2$  has 1024 (decimal). Calculate the value  $X_3$  stores after executing each line of the code sequence. All the data values are in decimal notation in Table 2. 2
- Answer in Table 1 below and you must show all your calculations in your script.

| Code Sequence         | Value of $X_3$ |
|-----------------------|----------------|
| <i>lw X3, 8(X1)</i>   |                |
| <i>slli X3, X3, 2</i> |                |
| <i>Add X3, X3, X2</i> |                |

Table - 1

| Memory Address | Data (in decimal) |
|----------------|-------------------|
| 0x00001232     | 10                |
| 0x00001236     | 12                |
| 0x0000123A     | 13                |
| 0x0000123E     | 14                |

Table - 2

CO2

- b. Given a machine code 2
- 0x00CB1A93
- i. Find the RISC-V assembly code of the given machine code.
- ii. Draw the datapath for the instruction you found in the previous question. You must mention the necessary control signals and proper labeling for all the components. 2

| Format | Instruction | Opcode  | Funct3 | Funct7/Funct 6 |
|--------|-------------|---------|--------|----------------|
| R      | ADD         | 0110011 | 001    | 0000000        |
|        | SUB         | 0110011 | 110    | 0000001        |
|        | AND         | 0110011 | 010    | 0000001        |
|        | OR          | 0110011 | 011    | 0000100        |
|        | SLL         | 0110011 | 101    | 0000100        |
| I      | LD          | 0000011 | 000    | N/A            |
|        | LW          | 0000011 | 010    | N/A            |
|        | LH          | 0000011 | 011    | N/A            |
|        | LB          | 0000011 | 001    | N/A            |
|        | ADDI        | 0010011 | 000    | N/A            |
|        | SLLI        | 0010011 | 001    | 000000         |
|        | SRLI        | 0010011 | 010    | 000000         |
|        | ORI         | 0010011 | 011    | N/A            |
| S      | SD          | 0100011 | 000    | N/A            |
|        | SW          | 0100011 | 001    | N/A            |
|        | SH          | 0100011 | 010    | N/A            |
|        | SB          | 0100011 | 011    | N/A            |

- c. Write the necessary RISC-V assembly instructions to compute the Euclidean distance: **3**

$$D = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2}$$

Finally, store the result in the memory location pointed to by x14.

The values a2, a1, b2, and b1 are stored in consecutive memory locations, each taking 4 bytes.

The starting address of a2 is in register x23 and D is mapped to register f15.

- d. What role does the offset play in base displacement addressing in RISC-V? What is the benefit of this system? **2**

3. a. Explain why the Immediate Generation Unit requires the entire instruction for processing, rather than just the 12 bits of the immediate field? **2**

CO3

b. Consider the code sequence given below.

```
ld x10, 32(x11)
add x5, x10, x7
addi x5, x5, 3
ld x13, 48(x5)
sd x13, 32(x5)
add x5, x13, x5
sub x6, x12, x5
sw x5, 12(x13)
sll x16, x6, x5
```

2

1

i. Considering a single cycle datapath, find out how many clock cycles are required for the above code sequence and **calculate** the average CPI?

2

ii. How many data hazards are there in the given code sequence if we run it in a single cycle datapath?

2

iii. How many data hazards are there in the given code sequence if we run it in a pipelined datapath?

2

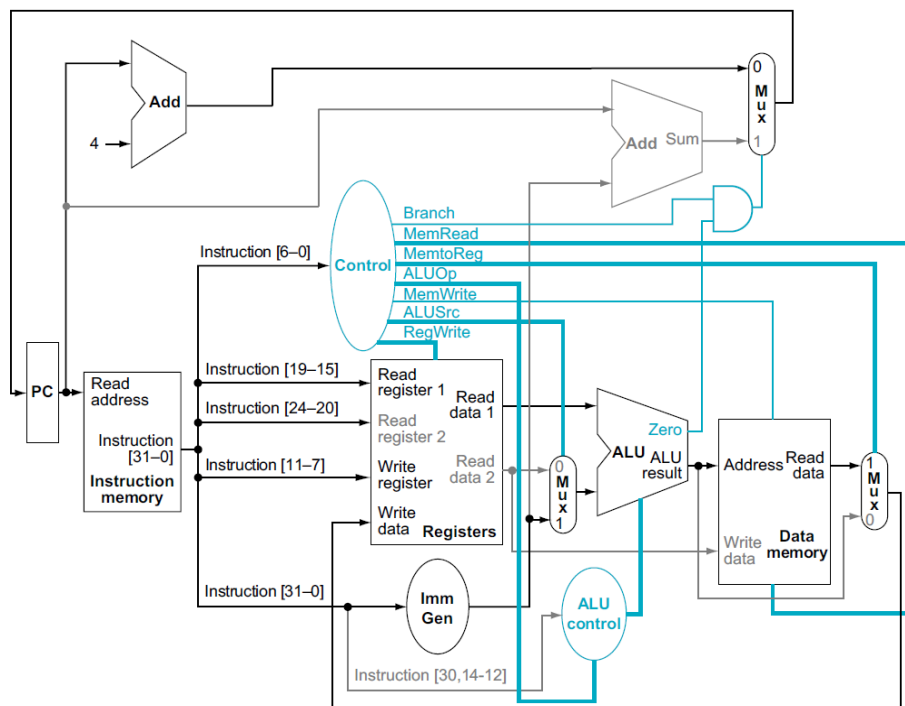
1

iv. Apply only **stall** to overcome the data hazards.

v. Apply only **forwarding** to overcome the data hazards.

vi. Calculate the total clock cycles and average CPI required after applying the method mentioned in the previous question(iii).

c.



Consider the above diagram. If you are executing the instruction **Bne X<sub>3</sub>, X<sub>4</sub>, Else**, what are the values of control signals generated by the control unit for this instruction?

Answer in the table below. For any unused control signal, write 'X'

3

| RegWrite | MemRead | MemWrite | ALUSrc | Branch | MemtoReg |
|----------|---------|----------|--------|--------|----------|
|          |         |          |        |        |          |

- |     |    |                                                                                                                                                                   |   |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | a. | Explain the significance of the offset field in an address with a proper example.                                                                                 | 1 |
| 4.  | b. | Explain why we need a memory hierarchy in our system.                                                                                                             | 2 |
|     | c. | Find the cache index, Tag, and offset bits for the memory address = 328, given that the Block size is 8 bits and the cache size is 32 bits.                       | 1 |
| CO4 | d. | "Choosing a large size for the cache line helps minimize the cache miss penalty." Justify the validity of this statement with appropriate reasoning and examples. | 2 |
|     | e. | Write a comparison between write-through and write-back techniques with proper examples.                                                                          | 1 |